

Customer Behaviour Analysis

Code link

https://github.com/Payal355-yadav/ALFIDO_Customer_Behavior_Analysis.ipynb.git

Objective

This project analyses customer purchasing behaviour to identify customer segments, spending patterns, and business opportunities using data analytics and machine learning techniques.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn (K-Means Clustering)

Methodology

1. Data Loading and Preprocessing
2. Data Cleaning (missing values and duplicates handling)
3. Feature Engineering (Total Spending calculation)
4. Customer Segmentation using K-Means Clustering
5. Data Visualization and Cluster Analysis
6. Business Recommendations

Key Analysis

- Customer purchase frequency was analysed.
- Total spending patterns were calculated.
- Customers were grouped into different clusters based on purchasing behaviour.
- Cluster profiles helped identify high-value and low-engagement customers.

Results

- Distinct customer segments were successfully identified.
- High-value customers can be targeted with loyalty programs.
- Low-engagement customers can be reactivated through personalized offers.
- Customer segmentation helps improve marketing efficiency and customer retention.

Conclusion

Customer Behaviour Analysis enables businesses to understand customer preferences, optimize marketing strategies, and improve overall customer satisfaction through data-driven decision making.